KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5431 | 5 Credits: 4 | Program Core | Source-grounded

Computer Integrated Design

 To introduce computer integration in all aspects of production system. To identify the features of CAM

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5431 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5431.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Mechatronics
Course code	5431
Course title	Computer Integrated Design
Semester	5 Credits: 4
Course category	Program Core
Periods per week	As specified in the official PDF

Item	Official information
Periods per semester	As specified in the official PDF
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

12 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- □ To introduce computer integration in all aspects of production system. □ To identify the features of CAM
- □ To provide an eye opening to the present scenario of a manufacturing plant so that the
- student can better understand their future job roles.
- □ To familiarize with the automation related to material handling, storage, inspection and manufacturing system

Course outcomes

CO	Official outcome	Study level
CO1	Familiarize the computer integration in design	See official syllabus
CO2	14 Understanding phase of product development. Identify the features of CAM 14 Understanding	See official syllabus
CO3	Describe the automation related to material handling,	See official syllabus
CO4	12 Understanding storage, inspection and manufacturing system.	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3

Row	Extracted official mapping
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Outline the automation opportunities and strategies.	3	As specified
Module 2	development.	2	As specified
Module 3	Identify the features of CAM.	4	As specified
Module 4	and manufacturing system.	3	As specified

MODULE 1

Outline the automation opportunities and strategies.

This module is presented from the official syllabus scope for Computer Integrated Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.

- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Outline the automation opportunities and strategies.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

2 Understanding in manufacturing system. Discuss about automation, its elements and

2 Understanding in manufacturing system. Discuss about automation, its elements and is an official topic in Module 1 of Computer Integrated Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding in manufacturing system. Discuss about automation, its elements and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding in manufacturing system. discuss about automation, its elements and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding in manufacturing system. Discuss about automation, its elements and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 2 Understanding in manufacturing system. Discuss about automation, its elements and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

12 Understanding types Discuss the principles and strategies for

12 Understanding types Discuss the principles and strategies for is an official topic in Module 1 of Computer Integrated Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 12 Understanding types Discuss the principles and strategies for using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 12 understanding types discuss the principles and strategies for should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 12 Understanding types Discuss the principles and strategies for means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-□□ 12 Understanding types Discuss the principles and strategies for □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

6 Understanding automation Concept of Computer Integrated Manufacturing (CIM); Opportunities for Automation and Computerization in a Production System– Facilities and Manufacturing Support Systems; Basic elements of automation Sensors – Desirable features, common types of sensors used in automation; Actuators - common types of actuators used in automation; Levels of automation; Types of Automation – Programmable Automation, Flexible Automation and Fixed Automation; Reasons for Automation; USA principle, Ten Strategies for Automation and Process Improvement; Automation Migration Strategy – Manual Production, Automated Production, Automated Integrated Production. Recognize the computer integration in design phase of product

6 Understanding automation Concept of Computer Integrated Manufacturing (CIM); Opportunities for Automation and Computerization in a Production System– Facilities and Manufacturing Support Systems; Basic elements of automation Sensors – Desirable features, common types of sensors used in automation; Actuators - common types of actuators used in automation; Levels of automation; Types of Automation – Programmable Automation, Flexible Automation and Fixed Automation; Reasons for Automation; USA principle, Ten Strategies for Automation and Process Improvement; Automation Migration Strategy – Manual Production, Automated Production, Automated Integrated Production. Recognize the computer integration in design phase of product is an official topic in Module 1 of Computer Integrated Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding automation Concept of Computer Integrated Manufacturing (CIM); Opportunities for Automation and Computerization in a Production System– Facilities and Manufacturing Support Systems; Basic elements of automation Sensors – Desirable features, common types of sensors used in automation; Actuators - common types of actuators used in automation; Levels of automation; Types of Automation – Programmable Automation, Flexible Automation and Fixed Automation; Reasons for Automation; USA principle, Ten Strategies for Automation and Process Improvement; Automation Migration Strategy – Manual

Production, Automated Production, Automated Integrated Production. Recognize the computer integration in design phase of product using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding automation concept of computer integrated manufacturing (cim); opportunities for automation and computerization in a production system– facilities and manufacturing support systems; basic elements of automation sensors – desirable features, common types of sensors used in automation; actuators - common types of actuators used in automation; levels of automation; types of automation – programmable automation, flexible automation and fixed automation; reasons for automation; usa principle, ten strategies for automation and process improvement; automation migration strategy – manual production, automated production, automated integrated production. recognize the computer integration in design phase of product should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding automation Concept of Computer Integrated Manufacturing (CIM); Opportunities for Automation and Computerization in a Production System– Facilities and Manufacturing Support Systems; Basic elements of automation Sensors – Desirable features, common types of sensors used in automation; Actuators - common types of actuators used in automation; Levels of automation; Types of Automation – Programmable Automation, Flexible Automation and Fixed Automation; Reasons for Automation; USA principle, Ten Strategies for Automation and Process Improvement; Automation Migration Strategy – Manual Production, Automated Production, Automated Integrated Production. Recognize the computer integration in design phase of product means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 6 Understanding automation Concept of Computer Integrated Manufacturing (CIM); Opportunities for Automation and Computerization in a Production System– Facilities and Manufacturing Support Systems; Basic elements of automation Sensors – Desirable features, common types of sensors used in automation; Actuators - common types of actuators used in automation; Levels of automation; Types of Automation – Programmable Automation, Flexible Automation and Fixed Automation; Reasons for Automation; USA principle, Ten Strategies for Automation and Process Improvement; Automation Migration Strategy – Manual Production, Automated Production, Automated Integrated Production. Recognize the computer integration in design phase of product □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Outline the automation opportunities and strategies.

             Identify the opportunities for computerization
   M1.01                                                          2
Understanding
             in manufacturing system.
             Discuss about automation, its elements and
   M1.02                                                          12
Understanding
             types
             Discuss the principles and strategies for
   M1.03                                                          6
Understanding
             automation
Contents:
Concept of Computer Integrated Manufacturing (CIM); Opportunities for Automation
and
Computerization in a Production System– Facilities and Manufacturing Support
Systems;

Basic elements of automation
Sensors – Desirable features, common types of sensors used in automation;
Actuators - common types of actuators used in automation;
Levels of automation; Types of Automation – Programmable Automation, Flexible
Automation
and Fixed Automation; Reasons for Automation;
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Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

development.

This module is presented from the official syllabus scope for Computer Integrated Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: development.
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

Describe about product development. 5 Understanding Describe the computer-aided designing and

Describe about product development. 5 Understanding Describe the computer-aided designing and is an official topic in Module 2 of Computer Integrated Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe about product development. 5 Understanding Describe the computer-aided designing and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe about product development. 5 understanding describe the computer-aided designing and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe about product development. 5 Understanding Describe the computer-aided designing and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Describe about product development. 5 Understanding Describe the computer-aided designing and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

9 Understanding prototyping. Product Development – Product Development Cycle and Product Development Spiral. Sequential and Concurrent Engineering (Introduction Level Only), Comparison. Computer Aided Design (CAD): Definition, Advantages, CAD hardware (Name and function) and software (Name Only); Geometric Modelling – Wireframe, Surface and Solid Modelling. Rapid Prototyping - Liquid Based, Solid Based and Powder Based Prototyping

9 Understanding prototyping. Product Development – Product Development Cycle and Product Development Spiral. Sequential and Concurrent Engineering (Introduction Level Only), Comparison. Computer Aided Design (CAD): Definition, Advantages, CAD hardware (Name and function) and software (Name Only); Geometric Modelling – Wireframe, Surface and Solid Modelling. Rapid Prototyping - Liquid Based, Solid Based and Powder Based Prototyping is an official topic in Module 2 of Computer Integrated Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 9 Understanding prototyping. Product Development – Product Development Cycle and Product Development Spiral. Sequential and Concurrent Engineering (Introduction Level Only), Comparison. Computer Aided Design (CAD): Definition, Advantages, CAD hardware (Name and function) and software (Name Only); Geometric Modelling – Wireframe, Surface and Solid Modelling. Rapid Prototyping - Liquid Based, Solid Based and Powder Based Prototyping using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 9 understanding prototyping. product development – product development cycle and product development spiral. sequential and concurrent engineering (introduction level only), comparison. computer aided design (cad): definition, advantages, cad hardware (name and function) and software (name only); geometric modelling – wireframe, surface and solid modelling. rapid prototyping - liquid based, solid based and powder based prototyping should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 9 Understanding prototyping. Product Development – Product Development Cycle and Product Development Spiral. Sequential and Concurrent Engineering (Introduction Level Only), Comparison. Computer Aided Design (CAD): Definition, Advantages, CAD hardware (Name and function) and software (Name Only); Geometric Modelling – Wireframe, Surface and Solid Modelling. Rapid Prototyping - Liquid Based, Solid Based and Powder Based Prototyping means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 9 Understanding prototyping. Product Development – Product Development Cycle and Product Development Spiral. Sequential and Concurrent Engineering (Introduction Level Only), Comparison. Computer Aided Design (CAD): Definition, Advantages, CAD hardware (Name and function) and software (Name Only); Geometric Modelling – Wireframe, Surface and Solid Modelling. Rapid Prototyping - Liquid Based, Solid Based and Powder Based Prototyping □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
development.

   M2.01      Describe about product development.                 5
Understanding
              Describe the computer-aided designing and
   M2.02                                                          9
Understanding
              prototyping.

              Series Test - I
Contents:
Product Development – Product Development Cycle and Product Development Spiral.

Sequential and Concurrent Engineering (Introduction Level Only), Comparison.

Computer Aided Design (CAD): Definition, Advantages, CAD hardware (Name and
function)
and software (Name Only); Geometric Modelling – Wireframe, Surface and Solid
Modelling.

Rapid Prototyping - Liquid Based, Solid Based and Powder Based Prototyping
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Identify the features of CAM.

This module is presented from the official syllabus scope for Computer Integrated Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified official hours	**CO-linked** see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Identify the features of CAM.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Describe CAM, CAPP 3 Remembering Explain the Variant and Generative techniques

Describe CAM, CAPP 3 Remembering Explain the Variant and Generative techniques is an official topic in Module 3 of Computer Integrated Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe CAM, CAPP 3 Remembering Explain the Variant and Generative techniques using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe cam, capp 3 remembering explain the variant and generative techniques should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe CAM, CAPP 3 Remembering Explain the Variant and Generative techniques means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Describe CAM, CAPP 3 Remembering Explain the Variant and Generative techniques □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

4 Understanding in CAPP Describe MPS, MRP, CRP and inventory

4 Understanding in CAPP Describe MPS, MRP, CRP and inventory is an official topic in Module 3 of Computer Integrated Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding in CAPP Describe MPS, MRP, CRP and inventory using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding in capp describe mps, mrp, crp and inventory should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding in CAPP Describe MPS, MRP, CRP and inventory means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-□□ 4 Understanding in CAPP Describe MPS, MRP, CRP and inventory □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

3 Remembering control with aid of computer.

3 Remembering control with aid of computer. is an official topic in Module 3 of Computer Integrated Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Remembering control with aid of computer. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 remembering control with aid of computer. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Remembering control with aid of computer. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 3 Remembering control with aid of computer. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Discuss about CNC Machines 4 Understanding Computer Aided Manufacturing (CAM)- Definition; Computer Aided Process Planning (CAPP)- Definition; Group Technology (Introduction Level treatment of cellular manufacturing, part family and its methods), Variant and Generative CAPP. Master Production Schedule (MPS), Material Requirements Planning (MRP), Capacity Requirement Planning (CRP) and Inventory Control with the aid of Computer (Introduction Level only). CNC and DNC- Features and Advantages; Machining Centres – Turning centre, Milling Centre. Discuss the automation related to material handling, storage, inspection

Discuss about CNC Machines 4 Understanding Computer Aided Manufacturing (CAM)- Definition; Computer Aided Process Planning (CAPP)- Definition; Group Technology (Introduction Level treatment of cellular manufacturing, part family and its methods), Variant and Generative CAPP. Master Production Schedule (MPS), Material Requirements Planning (MRP), Capacity Requirement Planning (CRP) and Inventory Control with the aid of Computer (Introduction Level only). CNC and DNC- Features and Advantages; Machining Centres – Turning centre, Milling Centre. Discuss the automation related to material handling, storage, inspection is an official topic in Module 3 of Computer Integrated Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss about CNC Machines 4 Understanding Computer Aided Manufacturing (CAM)- Definition; Computer Aided Process Planning (CAPP)- Definition; Group Technology (Introduction Level treatment of cellular manufacturing, part family and its methods), Variant and Generative CAPP. Master Production Schedule (MPS), Material Requirements Planning (MRP), Capacity Requirement Planning (CRP) and Inventory Control with the aid of Computer (Introduction Level only). CNC and DNC- Features and Advantages; Machining Centres – Turning centre, Milling Centre. Discuss the automation related to material handling, storage, inspection using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss about cnc machines 4 understanding computer aided manufacturing (cam)- definition; computer aided process planning (capp)- definition; group technology (introduction level treatment of cellular manufacturing, part family and its methods), variant and generative capp. master production schedule (mps), material requirements planning (mrp), capacity requirement planning (crp) and inventory control with the aid of computer (introduction level only). cnc and dnc- features and advantages; machining centres – turning centre, milling centre. discuss the automation related to material handling, storage, inspection should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discuss about CNC Machines 4 Understanding Computer Aided Manufacturing (CAM)- Definition; Computer Aided Process Planning (CAPP)- Definition; Group Technology (Introduction Level treatment of cellular manufacturing, part family and its methods), Variant and Generative CAPP. Master Production Schedule (MPS), Material Requirements Planning (MRP), Capacity Requirement Planning (CRP) and Inventory Control with the aid of Computer (Introduction Level only). CNC and DNC- Features and Advantages; Machining Centres – Turning centre, Milling Centre. Discuss the automation related to material handling, storage, inspection means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Discuss about CNC Machines 4 Understanding Computer Aided Manufacturing (CAM)- Definition; Computer Aided Process Planning (CAPP)- Definition; Group Technology (Introduction Level treatment of cellular manufacturing, part family and its methods), Variant and Generative CAPP. Master Production Schedule (MPS), Material Requirements Planning (MRP), Capacity Requirement Planning (CRP) and Inventory Control with the aid of Computer (Introduction Level only). CNC and DNC- Features and Advantages; Machining Centres – Turning centre, Milling Centre. Discuss the automation related to material handling, storage, inspection □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Identify the features of CAM.

   M3.01      Describe CAM, CAPP                                  3
Remembering


              Explain the Variant and Generative techniques
   M3.02                                                          4
Understanding
              in CAPP

              Describe MPS, MRP, CRP and inventory
   M3.03                                                          3
Remembering
              control with aid of computer.

   M3.04      Discuss about CNC Machines                          4
Understanding
Contents:
Computer Aided Manufacturing (CAM)- Definition;

Computer Aided Process Planning (CAPP)- Definition; Group Technology
(Introduction Level
treatment of cellular manufacturing, part family and its methods), Variant and
Generative
CAPP.

Master Production Schedule (MPS), Material Requirements Planning (MRP), Capacity
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

and manufacturing system.

This module is presented from the official syllabus scope for Computer Integrated Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: and manufacturing system.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

4 Understanding and storage systems Describe various automated inspection

4 Understanding and storage systems Describe various automated inspection is an official topic in Module 4 of Computer Integrated Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding and storage systems Describe various automated inspection using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding and storage systems describe various automated inspection should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding and storage systems Describe various automated inspection means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-□□ 4 Understanding and storage systems Describe various automated inspection □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

4 Understanding technologies. Describe the manufacturing systems and factors

4 Understanding technologies. Describe the manufacturing systems and factors is an official topic in Module 4 of Computer Integrated Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding technologies. Describe the manufacturing systems and factors using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding technologies. describe the manufacturing systems and factors should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding technologies. Describe the manufacturing systems and factors means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] 4 Understanding technologies. Describe the manufacturing systems and factors [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

4 Understanding for selection. Material handling – Fixed Routing and Variable Routing; List of Material Handling Equipments for Each Routing. Storage – Automated Storage/Retrieval System (AS/RS) and Carousel Storage System (Introduction Level Only). Inspection Technologies – Comparison of Contact and Non-Contact Inspection Techniques, CMM and Machine Vision (Introduction with Simple Diagram). Manufacturing Systems – Factors for Selection, Types – Single Station Cell, Multistation Fixed Routing, Multistation variable Routing (Introduction Level Only]. Text / Reference T/R Book Title/Author T1 Mikell P. Groover, Automation, Production Systems and Computer-Integrated Manufacturing. 4th edn. Pearson. T2 P.Radhakrishnan S.Subramanyan and V. Raju, CAD/CAM/CIM. 4th edn. New Age International Publishers. R1 S.R. Deb and S. Deb, Robotics Technology and Flexible Automation, Tata McGraw Hill. Online resources S.No Website Link 1 https://nptel.ac.in/courses/112/104/112104289/ 2 https://www.sciencedirect.com/search?qs=computer%20integrated%20manufactur ing

4 Understanding for selection. Material handling – Fixed Routing and Variable Routing; List of Material Handling Equipments for Each Routing. Storage – Automated Storage/Retrieval System (AS/RS) and Carousel Storage System (Introduction Level Only). Inspection Technologies – Comparison of Contact and Non-Contact Inspection Techniques, CMM and Machine Vision (Introduction with Simple Diagram). Manufacturing Systems – Factors for Selection, Types – Single Station Cell, Multistation Fixed Routing, Multistation variable Routing (Introduction Level Only]. Text / Reference T/R Book Title/Author T1 Mikell P. Groover, Automation, Production Systems and Computer-Integrated Manufacturing. 4th edn. Pearson. T2 P.Radhakrishnan S.Subramanyan and V. Raju, CAD/CAM/CIM. 4th edn. New Age International Publishers. R1 S.R. Deb and S. Deb, Robotics Technology and Flexible Automation, Tata McGraw Hill. Online resources S.No Website Link 1 https://nptel.ac.in/courses/112/104/112104289/ 2 https://www.sciencedirect.com/search?qs=computer%20integrated%20manufactur ing is an official topic in Module 4 of Computer Integrated Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding for selection. Material handling – Fixed Routing and Variable Routing; List of Material Handling Equipments for Each Routing. Storage – Automated Storage/Retrieval System (AS/RS) and Carousel Storage System (Introduction Level Only). Inspection Technologies – Comparison of Contact and Non-Contact Inspection Techniques, CMM and Machine Vision (Introduction with Simple Diagram). Manufacturing Systems – Factors for Selection, Types – Single Station Cell, Multistation Fixed Routing, Multistation variable Routing (Introduction Level Only]. Text / Reference T/R Book Title/Author T1 Mikell P. Groover, Automation, Production Systems and Computer-Integrated Manufacturing. 4th edn. Pearson. T2 P.Radhakrishnan S.Subramanyan and V. Raju, CAD/CAM/CIM. 4th edn. New Age International Publishers. R1 S.R. Deb and S. Deb, Robotics Technology and Flexible Automation, Tata McGraw Hill. Online resources S.No Website Link 1 https://nptel.ac.in/courses/112/104/112104289/ 2 https://www.sciencedirect.com/search?qs=computer%20integrated%20manufactur ing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding for selection. material handling – fixed routing and variable routing; list of material handling equipments for each routing. storage – automated storage/retrieval system (as/rs) and carousel storage system (introduction level only). inspection technologies – comparison of contact and non-contact inspection techniques, cmm and machine vision (introduction with simple diagram). manufacturing systems – factors for selection, types – single station cell, multistation fixed routing, multistation variable routing (introduction level only]. text / reference t/r book title/author t1 mikell p. groover, automation, production systems and computer-integrated manufacturing. 4th edn. pearson. t2 p.radhakrishnan s.subramanyan and v. raju, cad/cam/cim. 4th edn. new age international publishers. r1 s.r. deb and s. deb, robotics technology and flexible automation, tata mcgraw hill. online resources s.no website link 1 https://nptel.ac.in/courses/112/104/112104289/ 2 https://www.sciencedirect.com/search?qs=computer%20integrated%20manufactur ing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding for selection. Material handling – Fixed Routing and Variable Routing; List of Material Handling Equipments for Each Routing. Storage – Automated Storage/Retrieval System

Aspect	What to prepare
	(AS/RS) and Carousel Storage System (Introduction Level Only). Inspection Technologies – Comparison of Contact and Non-Contact Inspection Techniques, CMM and Machine Vision (Introduction with Simple Diagram). Manufacturing Systems – Factors for Selection, Types – Single Station Cell, Multistation Fixed Routing, Multistation variable Routing (Introduction Level Only]. Text / Reference T/R Book Title/Author T1 Mikell P. Groover, Automation, Production Systems and Computer-Integrated Manufacturing. 4th edn. Pearson. T2 P.Radhakrishnan S.Subramanyan and V. Raju, CAD/CAM/CIM. 4th edn. New Age International Publishers. R1 S.R. Deb and S. Deb, Robotics Technology and Flexible Automation, Tata McGraw Hill. Online resources S.No Website Link 1 https://nptel.ac.in/courses/112/104/112104289/ 2 https://www.sciencedirect.com/search?qs=computer%20integrated%20manufactur ing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 4 Understanding for selection. Material handling – Fixed Routing and Variable Routing; List of Material Handling Equipments for Each Routing. Storage – Automated Storage/Retrieval System (AS/RS) and Carousel Storage System (Introduction Level Only). Inspection Technologies – Comparison of Contact and Non-Contact Inspection Techniques, CMM and Machine Vision (Introduction with Simple Diagram). Manufacturing Systems – Factors for Selection, Types – Single Station Cell, Multistation Fixed Routing, Multistation variable Routing (Introduction Level Only]. Text / Reference T/R Book Title/Author T1 Mikell P. Groover, Automation, Production Systems and Computer-Integrated Manufacturing. 4th edn. Pearson. T2 P.Radhakrishnan S.Subramanyan and V. Raju, CAD/CAM/CIM. 4th edn. New Age International Publishers. R1 S.R. Deb and S. Deb, Robotics Technology and Flexible Automation, Tata McGraw Hill. Online resources S.No Website Link 1 https://nptel.ac.in/courses/112/104/112104289/ 2 https://www.sciencedirect.com/search?qs=computer%20integrated%20manufactur ing □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
and manufacturing system.
              Describe various automated material handling
   M4.01                                                              4
Understanding
              and storage systems
              Describe     various  automated     inspection
   M4.02                                                              4
Understanding
              technologies.
              Describe the manufacturing systems and factors
   M4.03                                                              4
Understanding
              for selection.
              Series Test - II
Contents:
Material handling – Fixed Routing and Variable Routing; List of Material Handling
Equipments for Each Routing.

Storage – Automated Storage/Retrieval System (AS/RS) and Carousel Storage System
(Introduction Level Only).

Inspection Technologies – Comparison of Contact and Non-Contact Inspection
Techniques,
CMM and Machine Vision (Introduction with Simple Diagram).

Manufacturing Systems – Factors for Selection, Types – Single Station Cell,
Multistation Fixed
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 2 Understanding in manufacturing system. Discuss about automation, its elements and. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding in manufacturing system. Discuss about automation, its elements and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 12 Understanding types Discuss the principles and strategies for. (3 marks)

Answer: Begin with the standard definition or identity of *12 Understanding types Discuss the principles and strategies for*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 6 Understanding automation Concept of Computer Integrated Manufacturing (CIM); Opportunities for Automation and Computerization in a Production System– Facilities and Manufacturing Support Systems; Basic elements of automation Sensors – Desirable features, common types of sensors used in automation; Actuators - common types of actuators used in automation; Levels of automation; Types of Automation – Programmable Automation, Flexible Automation and Fixed Automation; Reasons for Automation; USA principle, Ten Strategies for Automation and Process Improvement; Automation Migration Strategy – Manual Production, Automated Production, Automated Integrated Production. Recognize the computer integration in design phase of product. (3 marks)

Answer: Begin with the standard definition or identity of *6 Understanding automation Concept of Computer Integrated Manufacturing (CIM); Opportunities for Automation and Computerization in a Production System– Facilities and Manufacturing Support Systems; Basic elements of automation Sensors – Desirable features, common types of sensors used in automation; Actuators - common types of actuators used in automation; Levels of automation; Types of Automation – Programmable Automation, Flexible Automation and Fixed Automation; Reasons for Automation; USA principle, Ten Strategies for Automation and Process Improvement; Automation Migration Strategy – Manual Production, Automated Production, Automated Integrated Production. Recognize the computer integration in design phase of product*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain Describe about product development. 5 Understanding Describe the computer-aided designing and. (3 marks)

Answer: Begin with the standard definition or identity of *Describe about product development. 5 Understanding Describe the computer-aided designing and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 9 Understanding prototyping. Product Development – Product Development Cycle and Product Development Spiral. Sequential and Concurrent Engineering (Introduction Level Only), Comparison. Computer Aided Design (CAD): Definition, Advantages, CAD hardware (Name and function) and software (Name Only); Geometric Modelling – Wireframe, Surface and Solid Modelling. Rapid Prototyping - Liquid Based, Solid Based and Powder Based Prototyping. (3 marks)

Answer: Begin with the standard definition or identity of *9 Understanding prototyping. Product Development – Product Development Cycle and Product Development Spiral. Sequential and Concurrent Engineering (Introduction Level Only), Comparison. Computer Aided Design (CAD): Definition, Advantages, CAD hardware (Name and function) and software (Name Only); Geometric Modelling – Wireframe, Surface and Solid Modelling. Rapid Prototyping - Liquid Based, Solid Based and Powder Based Prototyping*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain Describe CAM, CAPP 3 Remembering Explain the Variant and Generative techniques. (3 marks)

Answer: Begin with the standard definition or identity of *Describe CAM, CAPP 3 Remembering Explain the Variant and Generative techniques*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding in CAPP Describe MPS, MRP, CRP and inventory. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding in CAPP Describe MPS, MRP, CRP and inventory*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Remembering control with aid of computer.. (3 marks)

Answer: Begin with the standard definition or identity of *3 Remembering control with aid of computer.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Discuss about CNC Machines 4 Understanding Computer Aided Manufacturing (CAM)- Definition; Computer Aided Process Planning (CAPP)- Definition; Group Technology (Introduction Level treatment of cellular

manufacturing, part family and its methods), Variant and Generative CAPP. Master Production Schedule (MPS), Material Requirements Planning (MRP), Capacity Requirement Planning (CRP) and Inventory Control with the aid of Computer (Introduction Level only). CNC and DNC- Features and Advantages; Machining Centres – Turning centre, Milling Centre. Discuss the automation related to material handling, storage, inspection. (3 marks)

Answer: Begin with the standard definition or identity of *Discuss about CNC Machines 4 Understanding Computer Aided Manufacturing (CAM)- Definition; Computer Aided Process Planning (CAPP)- Definition; Group Technology (Introduction Level treatment of cellular manufacturing, part family and its methods), Variant and Generative CAPP. Master Production Schedule (MPS), Material Requirements Planning (MRP), Capacity Requirement Planning (CRP) and Inventory Control with the aid of Computer (Introduction Level only). CNC and DNC- Features and Advantages; Machining Centres – Turning centre, Milling Centre. Discuss the automation related to material handling, storage, inspection*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain 4 Understanding and storage systems Describe various automated inspection. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding and storage systems Describe various automated inspection*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding technologies. Describe the manufacturing systems and factors. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding technologies. Describe the manufacturing systems and factors*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 4 Understanding for selection. Material handling – Fixed Routing and Variable Routing; List of Material Handling Equipments for Each Routing. Storage – Automated Storage/Retrieval System (AS/RS) and Carousel Storage System (Introduction Level Only). Inspection Technologies – Comparison of Contact and Non-Contact Inspection Techniques, CMM and Machine Vision (Introduction with Simple Diagram). Manufacturing Systems – Factors for Selection, Types – Single Station Cell, Multistation Fixed Routing, Multistation variable Routing (Introduction Level Only]. Text / Reference T/R Book Title/Author T1 Mikell P. Groover, Automation, Production Systems and Computer-Integrated Manufacturing. 4th edn. Pearson. T2 P.Radhakrishnan S.Subramanyan and V. Raju, CAD/CAM/CIM. 4th edn. New Age International Publishers. R1 S.R. Deb and S. Deb, Robotics Technology and Flexible Automation, Tata McGraw Hill. Online resources S.No Website Link 1 https://nptel.ac.in/courses/112/104/112104289/ 2 https://www.sciencedirect.com/search?qs=computer%20integrated%20manufactur ing. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding for selection. Material handling – Fixed Routing and Variable Routing; List of Material Handling Equipments for Each Routing. Storage – Automated Storage/Retrieval System (AS/RS) and Carousel Storage System (Introduction Level Only). Inspection Technologies – Comparison of Contact and Non-Contact Inspection Techniques, CMM and Machine Vision (Introduction with Simple Diagram). Manufacturing Systems – Factors for Selection, Types – Single Station Cell, Multistation Fixed Routing, Multistation variable Routing (Introduction Level Only]. Text / Reference T/R Book Title/Author T1 Mikell P. Groover, Automation, Production Systems and Computer-Integrated Manufacturing. 4th edn. Pearson. T2 P.Radhakrishnan S.Subramanyan and V. Raju, CAD/CAM/CIM. 4th edn. New Age International Publishers. R1 S.R. Deb and S. Deb, Robotics Technology and Flexible Automation, Tata McGraw Hill. Online resources S.No Website Link 1 https://nptel.ac.in/courses/112/104/112104289/ 2 https://www.sciencedirect.com/search?qs=computer%20integrated%20manufactur ing*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **2 Understanding in manufacturing system. Discuss about automation, its elements and**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Describe about product development. 5 Understanding Describe the computer-aided designing and**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Describe CAM, CAPP 3 Remembering Explain the Variant and Generative techniques**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **4 Understanding and storage systems Describe various automated inspection**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- S.No Website Link
- https://nptel.ac.in/courses/112/104/112104289/
- https://www.sciencedirect.com/search?qs=computer%20integrated%20manufactur

IN-PAGE SEARCH

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